

EX.NO:3 Exploring the different types of prompts- Straightforward Prompts - Tabular Format Prompting - Missing Word Prompting - Preceding Question Prompting.

Name: **joel benedict AP**

Reg.no: 212225050019

Subject: Prompt Engineering

1. Straightforward Prompts

Scenario 1 – Education

Prompt: Explain the concept of blockchain technology in 150 words using simple language suitable for beginners.

Output:

Blockchain is a digital system used to store information in a secure and transparent way. It works like a chain of blocks where each block contains data, a timestamp, and a unique code that links it to the previous block. Once information is added to a block, it cannot be easily changed, making the system very secure. Blockchain technology does not rely on a central authority such as a bank or government. Instead, it is maintained by a network of computers called nodes that verify and record transactions. This decentralized structure helps prevent fraud and improves trust between users. Blockchain is mainly used in cryptocurrencies like Bitcoin, but it is also used in supply chain management, healthcare, and digital contracts. Because of its transparency, security, and reliability, blockchain technology is becoming an important innovation in modern digital systems.

Scenario 2 – Technology

Prompt: Write a 200-word article describing the advantages and disadvantages of artificial intelligence in healthcare.

Output:

Artificial intelligence (AI) is transforming the healthcare industry by improving diagnosis, treatment, and patient care. One of the major advantages of AI is its ability to analyze large amounts of medical data quickly and accurately. AI systems can assist doctors in detecting diseases such as cancer at an early stage, which increases the chances of successful treatment. AI-powered tools can also help hospitals manage patient records, predict disease outbreaks, and support medical research.

Another benefit is automation. AI can handle repetitive tasks such as scheduling appointments or monitoring patient conditions, allowing healthcare professionals to

focus more on patient care. It also improves decision-making by providing data-driven insights.

However, AI in healthcare also has disadvantages. High implementation costs make it difficult for smaller hospitals to adopt advanced AI technologies. There are also concerns about data privacy and security because patient information is highly sensitive. In addition, AI systems depend heavily on the quality of the data they are trained on; incorrect or biased data may lead to inaccurate results.

Overall, while AI offers many benefits, careful regulation and ethical use are necessary to ensure it supports healthcare professionals effectively.

2. Tabular Format Prompting

Scenario 1 – Prompting Techniques

Prompt:

“Create a well-structured table comparing different prompting techniques: Zero-Shot Prompting, Few-Shot Prompting, Chain-of-Thought Prompting, and Role-Based Prompting.

The table should include the following columns:

- Prompting Technique**
- Definition**
- Key Features**
- Advantages**
- Example Use Case**

Prompting Technique	Definition	Key Features	Advantages	Example Use Case
Zero-Shot Prompting	Asking the AI to perform a task without examples	Direct instruction	Quick and simple	Text summarization
Few-Shot Prompting	Providing a few examples before the task	Uses sample inputs and outputs	Improves accuracy	Sentiment analysis

Prompting Technique	Definition	Key Features	Advantages	Example Use Case
Chain-of-Thought Prompting	Encourages step-by-step reasoning	Logical explanation	Better problem solving	Mathematical problems
Role-Based Prompting	Assigning a specific role to the AI	AI acts as an expert	More contextual responses	Expert consultation

Scenario 2 – Programming Languages Comparison

Prompt:

“Create a neatly formatted comparison table for three programming languages: Python, Java, and C++.

The table should include the following columns:

- **Language**
- **Primary Use**
- **Advantages**
- **Disadvantages**
- **Popular Applications**

Language	Primary Use	Advantages	Disadvantages	Popular Applications
Python	Data science, AI, web development	Easy to learn, large libraries	Slower execution speed	Machine learning, automation
Java	Enterprise software, Android apps	Platform independent	Requires more memory	Mobile apps, enterprise systems
C++	System programming, game development	High performance	Complex syntax	Game engines, operating systems

3. Preceding Question Prompting

Scenario 1 – Climate Change

What is your current lifestyle?

(urban, rural, student, working professional)

How do you usually commute?

(public transport, bike, car)

Are you willing to invest money in eco-friendly solutions (like solar panels)?

Prompt Question: What are the major causes of climate change?

Answer:

The major causes of climate change include the burning of fossil fuels, deforestation, industrial emissions, and agricultural activities. These activities release greenhouse gases such as carbon dioxide and methane into the atmosphere.

Explanation:

Climate change has a significant impact on global ecosystems. Rising temperatures cause glaciers and polar ice caps to melt, leading to higher sea levels and coastal flooding. Changes in temperature and rainfall patterns also affect agriculture and water availability. Many plant and animal species struggle to adapt to these environmental changes, which can lead to biodiversity loss. Climate change also increases the frequency of extreme weather events such as storms, droughts, and heatwaves. These impacts affect not only the environment but also human health and economies around the world. Addressing climate change requires reducing greenhouse gas emissions and adopting sustainable energy sources.

Scenario 2 – Cybersecurity

What devices do you use regularly? (mobile, laptop, both)

How do you currently manage your passwords? (same password, password manager, etc.)

Have you ever experienced hacking or phishing attacks?

Prompt Question: Why is cybersecurity important in modern digital systems?

Answer:

Cybersecurity is important because it protects computer systems, networks, and data from cyberattacks, unauthorized access, and data breaches.

Best Practices:

1. Use strong and unique passwords.
 2. Enable two-factor authentication.
 3. Keep software and operating systems updated.
 4. Avoid clicking suspicious links or emails.
 5. Use reliable antivirus and security tools.
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4. Missing Word Prompting

Scenario 1 – Science

Prompt:

Photosynthesis is the process by which _____ convert _____ into chemical energy using sunlight.

Output:

Photosynthesis is the process by which **plants** convert **carbon dioxide and water** into chemical energy using sunlight.

Scenario 2 – Computer Networks

Prompt:

The protocol used to transfer web pages over the internet is _____, and the secure version of this protocol is _____.

Output:

The protocol used to transfer web pages over the internet is **Hypertext Transfer Protocol**, and the secure version of this protocol is **HTTPS**.